

Global Facilities - Design & Construction - General - Design Parameters Requirements

ECN Facility:	GLOBAL FACILITIES	ECN Area:	Engineering
Owner(s):	Engineering	Target Audience:	All Facilities
Document Type:	Standard	Document ID: (pdf)	
Version:	4		ZHNT2QEMQUW-1202974420-161
Current ECN:	101141620	Last Modified:	04/20/2023

[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 The purpose of this document is to define appropriate system design requirements as Basis of Design (BOD) for new Greenfield Construction FABs, including Frontend Manufacturing and Backend Assembly & Test (A&T) sites.
- 1.2 This document is limited to the initial design of the building facility systems, and/or to transfer manufacturing processes to an existing FAB being converted to the Micron manufacturing process.
- 1.3 Once the initial building construction or remodel construction is completed, and the facility system installations are completed, certified, and/or re-certified, this document no longer applies. At that point, please refer to individual System BKM Documents or Global System Standards.

2.0 Scope

- 2.1 System(s) impacted
 - 2.1.1 This document applies to all Facilities systems outlined in the document.
- 2.2 Site(s) and Person(s) impacted
 - 2.2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites, including Frontend Manufacturing and Backend Assembly & Test (A&T) sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
BKM	Best Known Method
BOD	Basis of Design
EOL	End of Line
FAC	Facilities
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
POC	Point of Connection
POD	Point of Delivery
POU	Point of Use

4.0 Roles and Responsibilities

Roles	Responsibilities
Global FAC Engineering (Global Team)	• Develop and maintain document • Ensure Best Known Methods (BKM) and lessons learned are captured in document • Perform periodic document review and update document
Global FAC – Regional Engineering	• Review and provide technical feedback on document • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on document • Incorporate document into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from document • Share new lessons learned to Global Facilities - Global Engineering and Regional Engineering teams, to determine if they are BKMs that can be captured in future revision of document
Site Facilities Teams	• Use document as technical reference • Share new lessons learned to Global Facilities - Global Engineering and Regional Engineering teams, to determine if they are BKMs that can be captured in future revision of document

5.0 General Requirements

Appendix A to T show the relevant design parameters for all Micron sites to refer to.

- 5.1 Unless otherwise noted, all values are defined at the Point of Connection (POC) at the tool.
- 5.2 Points of connection may also be referred to as: Point of Delivery (POD), Point of Use (POU), or similar acronyms utilized at each site.
- 5.3 Point of connection is located at the tool boundary, in the space where the utilities are physically connected to the tool frame, or its sub-equipment. Parameters shall be measured as specified in the individual system BKM document or Global System Standards.
- 5.4 Site to evaluate existing site specification in comparison to the design parameters; should there's any discrepancy, evaluation to be done to follow the stringent specification to meet Process requirements.

SN	Design Parameters	Appendix
1	Bulk Gases	Appendix A
2	Specialty Gases	Appendix B
3	City Water	Appendix C
4	Cleanroom Environment	Appendix D
5	Cleanroom Noise	Appendix E
6	Cleanroom Particles	Appendix F
7	Cleanroom Vibration	Appendix G
8	Chemicals	Appendix H
9	Slurries	Appendix I
10	Compressed Air (CDA/HPCDA)	Appendix J
11	Electrical	Appendix K
12	Electrostatic Discharge	Appendix L
13	Exhaust	Appendix M
14	Fab Height & Load Limitations	Appendix N
15	Freight Elevator	Appendix O
16	Install Dock for Heavy Equipment	Appendix P
17	Process Cooling Water	Appendix Q
18	Vacuum	Appendix R
19	Ultra Pure Water/Reclaim Water	Appendix S
20	Waste Drains	Appendix T

6.0 Document Control

6.1 **Approval:** GLOBAL_FAC_GFTT_APPR

6.2 **Notification:**

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_EHS_TEAM_MEMBERS
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

6.3 **Final Viewer Access**

All Micron sites Facilities Team Members

6.4 **Retention**

Adherence to the requirements specified in the

[Global Facilities - Document Control Procedures](#)

6.5 **Review**

Biennially (two years)

7.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	31/03/2020
1	Main Page Whole document	Name change from "BKM Design Parameters" to "Global Facilities - General - Design Parameters BKM" Format change from BKM to System Standard Changes throughout whole document	05/27/2020
2	Title Appendix D and F	Changed title from "Global Facilities - General - Design Parameters Best Known Methods (BKM)" to "Global Facilities - Design & Construction - General - Design Parameters Requirements" Added note for Backend Assembly and Test (A&T) sites to refer to linked document for Cleanroom Temperature, Humidity, and Particle Specs	02/16/2022
3	3.0 Appendix D Appendix F Appendix G Appendix J Appendix N	Added Term/Acronym related to reorganization: FCT, GFTT, GFC, Global Facilities Central Team, Global Facilities Construction Team, Global Facilities Operations Support Team, Global Facilities Planning Team. Added Section 3 for Probe Temperature & Humidity Spec Added Sections 2, 4, and 5 for Reticle Room, Probe Cleanroom, and Probe Support Area - Cleanroom Particle Specs Added Sections 2 and 3 for FAB Remainder Cleanroom and Probe Vibration Specs Modified Sections 1b and 2b for CDA and HPCDA Dewpoint Spec tolerance to "<" Added Section 8 for FAB Cleanroom Floor Tile (Heavy Duty Type) Concentrated Load	04/20/2023

Appendix A: Design Parameters for Bulk Gases

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Process Nitrogen				
a	Supply Pressure	≥ 105	N/A	psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust the pressure requirement accordingly
b	O ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
c	H ₂ O	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
d	CO	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
e	CO ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
f	H ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
g	THC (expressed as CH ₄)	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
h	Particles	5 max, 1 avg	N/A	# particles (>0.1µm)/scf	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas or by Pressurized Gas Cylinder" for test method.
2	General Nitrogen				
a	Supply Pressure	≥ 105	N/A	psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust the pressure requirement accordingly
b	O ₂	≤ 100	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
c	H ₂ O	≤ 100	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
d	CO	≤ 100	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
e	CO ₂	≤ 100	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
f	H ₂	≤ 100	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
g	THC (expressed as CH ₄)	≤ 100	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
3	Utility Nitrogen				
a	Supply Pressure	≥ 105	N/A	psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust the pressure requirement accordingly
b	O ₂	≤ 2000	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
c	H ₂ O	≤ 2000	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
d	H ₂	≤ 2000	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.

Appendix A: Design Parameters for Bulk Gases (Continued)

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
4	Process Oxygen				
a	Supply Pressure	≥ 65	N/A	psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust the pressure requirement accordingly
b	Ar	≤ 50	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
c	N ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
d	H ₂ O	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
e	CO	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
f	CO ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
g	H ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
h	THC (expressed as CH ₄)	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
i	Particles	5 max, 1 avg	N/A	# particles (>0.1µm)/scf	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas or by Pressurized Gas Cylinder" for test method.
5	Process Hydrogen				
a	Supply Pressure	≥ 65	N/A	psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust the pressure requirement accordingly
b	N ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
c	O ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
d	H ₂ O	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
e	CO	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
f	CO ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
g	THC (expressed as CH ₄)	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
h	Particles	5 max, 1 avg	N/A	# particles (>0.1µm)/scf	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas or by Pressurized Gas Cylinder" for test method.
6	Process Argon				
a	Supply Pressure	≥ 65	N/A	psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust the pressure requirement accordingly
b	N ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
c	O ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
d	H ₂ O	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
e	CO	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
f	CO ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
g	H ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
h	THC (expressed as CH ₄)	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
i	Particles	5 max, 1 avg	N/A	# particles (>0.1µm)/scf	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas or by Pressurized Gas Cylinder" for test method.

Appendix A: Design Parameters for Bulk Gases (Continued)

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
7	Process Helium				
a	Supply Pressure	≥ 65	N/A	psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust the pressure requirement accordingly
b	O ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
c	H ₂ O	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
d	CO	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
e	CO ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
f	H ₂	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
g	THC (expressed as CH ₄)	≤ 10	N/A	ppb	Maximum concentration as measured at the point of supply as for system design specification.
h	Particles	5 max, 1 avg	N/A	# particles (>0.1µm)/scf	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas or by Pressurized Gas Cylinder" for test method.

Appendix B: Design Parameters for Specialty Gases

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Bulk & Cylinder Specialty Gases	≥ 70	See note	psig	All positive pressure gases, having a gas source pressure between 80 and 2460 Psig
2	Low Vapor Pressure Gases	See note	See note	psig	All gases having vapor pressure between 4.3 and less than 80 Psig, according to tool requirements

Appendix C: Design Parameters for City Water

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
a	N/A	N/A	N/A	N/A	

Appendix D: Design Parameters for Cleanroom Environment

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	FAB - Photo /Lithography Areas including EUV and Reticle Room				
a	Temperature	72.5	± 1.8	° F	Temperature Spec is equivalent to 22.5 ± 1.0 °C. New FAB Design should be based on the eventual Cleanroom Humidity Spec from Global Spec Alignment project: 42 ± 3% (in the future). Current Global Spec Alignment values is DRAM: 43 ± 3%, NAND: 45 ± 5%.
b	Humidity	42	± 3	%	
2	FAB - Remainder of Cleanroom Ballroom Areas				
a	Temperature	72.5	± 1.8	° F	Temperature Spec is equivalent to 22.5 ± 1.0 °C. New FAB Design should be based on the eventual Cleanroom Humidity Spec from Global Spec Alignment project: 42 ± 3% (in the future). Current Global Spec Alignment values is DRAM: 43 ± 3%, NAND: 45 ± 5%.
b	Humidity	42	± 3	%	
3	Probe Cleanroom Areas				
a	Temperature	73.4	± 5.4	° F	Temperature Spec is equivalent to 23.0 ± 3.0 °C
b	Humidity	45	± 5	%	
4	Non-Critical Cleanroom/Support Areas (AMHS Link Bridge, Gowning Room, Clean Chem Lab, Decon Room, Quartz Cleaning)				
a	Temperature	72.5	± 3.6	° F	Temperature Spec is equivalent to 22.5 ± 2.0 °C
b	Humidity	42	± 3	%	New FAB Design should be based on the eventual Cleanroom Humidity Spec from Global Spec Alignment project: 42 ± 3% (in the future). Current Global Spec Alignment values is DRAM: 43 ± 3%, NAND: 45 ± 5%.

***Note:** For Cleanroom Temperature and Humidity Specs of Backend Assembly and Test (A&T) sites, refer to the "Global AT – Cleanroom ISO 8 and Below – Operations, Maintenance, and Controls (EN)" document.

Appendix E: Design Parameters for Cleanroom Noise

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Photo Lithography / Metrology Areas				
a	Noise Level	60	N/A	NC	As-built condition for Photolithography area with tools installed
2	Remainder of Cleanroom Ballroom				
a	Noise Level	65	N/A	NC	As-built condition for Production Clean Room area with tools installed

Appendix F: Design Parameters for Cleanroom Particles

Section	Description	FOR GLOBAL BOD			Notes/Comments
		ISO Classification	Tolerance	Units	
1	FAB Production Cleanroom Areas (Photo Lithography Areas including EUV/Remainder Cleanroom Ballroom)	ISO Class 5 (Turbulent)	N/A	N/A	
2	FAB Reticle Room	ISO Class 3 (Turbulent)	N/A	N/A	

Last Modified: 4/30/23

Version: 4

ID: ZHNTE2QEMQUW-1202974420-161

3	FAB - Non-Critical Cleanroom/Support Areas (AMHS Link Bridge, Gowning Room, Clean Chem Lab, Quart Cleaning)	ISO Class 6 (Turbulent)	N/A	N/A	
4	Probe Cleanroom Areas (Probe)	ISO Class 6 (Turbulent)	N/A	N/A	
5	Probe - Non-Critical Cleanroom/Support Areas (Probe Gowning Room, Probe Decon Room)	ISO Class 7 (Turbulent)	N/A	N/A	

***Note:** For Cleanroom Particle Specs of Backend Assembly and Test (A&T) sites, refer to the "[Global AT – Cleanroom ISO 8 and Below – Operations, Maintenance, and Controls \(EN\)](#)" document.

Appendix G: Design Parameters for Cleanroom Vibration

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	FAB - Photo Lithography Areas including EUV, and Metrology/RDA Areas – FAB Level Waffle Slab	200	N/A	μ-in/sec	Modified VC-D+ scale. Equivalent to 5.08 μ-m/sec As-built condition after turn over to owner, before tools are installed.
2	FAB – All Remainder Cleanroom Areas except Photo/EUV/Metrology/RDA – FAB Level Waffle Slab	500	N/A	μ-in/sec	VC-C scale. Equivalent to ~12.5 μ-m/sec As-built condition after turn over to owner, before tools are installed.
3	Probe Cleanroom – Cleanroom Level Slab	500	N/A	μ-in/sec	VC-C scale. Equivalent to ~12.5 μ-m/sec As-built condition after turn over to owner, before tools are installed.

Appendix H: Design Parameters for Chemicals

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	CMP Chemicals	Metal (outlet-inlet) ≤ 0.1 ppb (0.1μm) particle ≤ 10 pcs/ml		(ppb pcs/ml)	
a	Supply Pressure minimum @ POC	35	± 3	psig	All system temps are ambient based on building/room temp specs, however this may change depending on chemical delivery engine type and process requirements. High viscosity chemicals may be agreed at lower delivery pressure (i.e. H2SO4. H3PO4). Note 1: Minimum pressure is based on typical tool vendor requirements. Several factors influence the ability to achieve this value and in some cases this value may need to be adjusted based on the design criteria. System pressure shall not exceed the maximum manufactures recommended operating pressure of any system component.

Appendix I: Design Parameters for Slurries

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	

Last Modified: 4/30/23

Version: 4

ID: ZHNTE2QEMQUW-1202974420-161

1	CMP Chemicals	Metal (outlet-inlet) ≤ 0.1 ppb (0.1 μ m) particle ≤ 10 pcs/ml		(ppb pcs/ml)	
a	Supply Pressure minimum @ POC	30	± 3	psig	Temperature to be kept at building spec. However this may change depending on chemical delivery engine type and process requirements. Consult Process owners to derive critical temperature parameters for each slurry type delivered. Note 1: Minimum pressure is based on typical tool vendor requirements. Several factors influence the ability to achieve this value and in some cases this value may need to be adjusted based on the design criteria. System pressure shall not exceed the maximum manufactures recommended operating pressure of any system component.
b	Velocities minimum in loop	2.1	± 1	fps	Note 2: Minimum velocity considerations; slurry properties, particle suspension characteristics, slurry manufacture recommendations and established best practices. This value should be adjusted where required.

Appendix J: Design Parameters for Compressed Air (CDA/HPCDA)

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Clean Dry Air (CDA)				
a	Supply Pressure	116	± 4	psig	Pressure Spec is equivalent to 0.80 ± .027 MPa. Measured at any single end point or the most hydraulically remote end-of-line.
b	Dewpoint (Pressure Dewpoint)	-40	<	° F	Based on Pressure Dew Point. Equivalent to < -40 °C.
2	High Pressure Clean Dry Air (HPCDA)				
a	Supply Pressure	138	± 4	psig	Pressure Spec is equivalent to 0.951 ± .027 MPa. Measured at any single end point or the most hydraulically remote end-of-line.
b	Dew point (Pressure Dewpoint)	-40	<	° F	Based on Pressure Dew Point. Equivalent to < -40 °C.

Appendix K: Design Parameters for Electrical**i) For US**

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Voltage Tolerance	Frequency Tolerance (Hz)	Units	
1	480V – 3 Phase – 60 Hz	+ 5 / -5	± 0.5	%	Tolerances shall be maintained with a SAG Mitigation solution or SEMI F47 compliant tools
2	480V – 3 Phase – 60 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
3	277V – 1 Phase – 60 Hz	+ 5 / -5	± 0.5	%	Tolerances shall be maintained with a SAG Mitigation solution or SEMI F47 compliant tools
4	208V – 3 Phase – 60 Hz	+ 5 / -5	± 0.5	%	Tolerances shall be maintained with a SAG Mitigation solution or SEMI F47 compliant tools
5	208V – 3 Phase – 60 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
6	208V – 1 Phase – 60 Hz	+ 5 / -5	± 0.5	%	Tolerances shall be maintained with a SAG Mitigation solution or SEMI F47 compliant tools
7	208V – 1 Phase – 60 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
8	120V – 1 Phase – 60 Hz	+ 5 / -5	± 0.5	%	Tolerances shall be maintained with a SAG Mitigation solution or SEMI F47 compliant tools
9	120V – 1 Phase – 60 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	

Appendix K: Design Parameters for Electrical**ii) For Europe / Asia (Singapore, Malaysia)**

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Voltage Tolerance	Frequency Tolerance (Hz)	Units	
1	480V – 3 Phase – 50 Hz	+ 5 / -5	± 0.5	%	Tolerances for critical loads shall be maintained with a SAG Mitigation solution or SEMI F47 compliant equipment.
2	480V – 3 Phase – 50 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
3	400V – 3 Phase – 50 Hz	+ 5 / -5	± 0.5	%	Tolerances for critical loads shall be maintained with a SAG Mitigation solution or SEMI F47 compliant equipment.
4	400V – 3 Phase – 50 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
5	230V – 1 Phase – 50 Hz	+ 5 / -5	± 0.5	%	Tolerances for critical loads shall be maintained with a SAG Mitigation solution or SEMI F47 compliant equipment.
6	230V – 1 Phase – 50 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
7	208V – 3 Phase – 50 Hz	+ 5 / -5	± 0.5	%	Tolerances for critical loads shall be maintained with a SAG Mitigation solution or SEMI F47 compliant equipment.
8	208V – 3 Phase – 50 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
9	208V – 1 Phase – 50 Hz	+ 5 / -5	± 0.5	%	Tolerances for critical loads shall be maintained with a SAG Mitigation solution or SEMI F47 compliant equipment.
10	208V – 1 Phase – 50 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	
11	120V – 1 Phase – 50 Hz	+ 5 / -5	± 0.5	%	Tolerances for critical loads shall be maintained with a SAG Mitigation solution or SEMI F47 compliant equipment.
12	120V – 1 Phase – 50 Hz (UPS)	+ 1.5 / -1.5	± 0.1	%	

Appendix L: Design Parameters for Electrostatic Discharge

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Limits	1E5 - 1E8	N/A	Ohms	

Appendix M: Design Parameters for Exhaust

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Acid Exhaust (AE)				
a	Pressure	-3.6	± 0.2	in. W.C.	Pressure Spec is equivalent to -900 ± 50 Pa. Measured at any single end point or the most hydraulically remote end-of-line.
2	Ammonia Exhaust (AME)				
a	Pressure	-3.6	± 0.2	in. W.C.	Pressure Spec is equivalent to -900 ± 50 Pa. Measured at any single end point or the most hydraulically remote end-of-line.
3	Solvent Exhaust (SE)				
a	Pressure	-3.6	± 0.2	in. W.C.	Pressure Spec is equivalent to -900 ± 50 Pa. Measured at any single end point or the most hydraulically remote end-of-line.
4	General Exhaust (GE)				
a	Pressure	-3.0	± 0.2	in. W.C.	Pressure Spec is equivalent to -750 ± 50 Pa. Measured at any single end point or the most hydraulically remote end-of-line.

Appendix N: Design Parameters for Fab Height & Load Limitations

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Ceiling (Fab Level) - Single Level AMHS	192	N/A	in.	Equivalent to 4.88m height. Above raised floor. To prevent under track storage to move in largest pieces of equipment.
2	Ceiling (Fab Level) - Double Level AMHS	255.9	N/A	in.	Equivalent to 6.5m height. Above raised floor. To prevent under track storage to move in largest pieces of equipment.
3	Subfab Move In Height	108	N/A	in.	To allow for facilitation of subfab equipment. This is to bottom of racking.
4	Sprinkler Height (FAB Level)	N/A	N/A	in.	Depends on FAB Ceiling Height
5	AMHS Under Track Storage (UTS)	N/A	N/A	in.	Depends on FAB Ceiling Height. Lighting is side track storage to allow for work around the lowest point of work near under track storage.
6	FAB Cleanroom Floor Tile Concentrated Load – Perimeter Aisle Move-In Path	3,000	N/A	lbf/in2	Equivalent to ~1,400 kgf/in2.
7	FAB Cleanroom Floor Tile Concentrated Load – Waffle Floor Production Area	3,000	N/A	lbf/in2	Equivalent to ~1,400 kgf/in2.
8	FAB Cleanroom Floor Tile (Heavy Duty Type) Concentrated Load – FAB Waffle Floor Production Area	6,600	N/A	lbf/in2	Equivalent to ~3,000 kgf/in2.

Appendix O: Design Parameters for Freight Elevator

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Door Height	120	N/A	in.	
2	Door Width	128	N/A	in.	
3	Depth	186	N/A	in.	
4	Load Capacity (Max)	30,000	N/A	lbs.	Reference Only. To be verified upon design of new facility

Appendix P: Design Parameters for Install Dock for Heavy Equipment

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Platform Base Width	96	N/A	in.	Reference Only. To be verified upon design of new facility
2	Platform Base Length	144	N/A	in.	Reference Only. To be verified upon design of new facility
3	Platform Lifting Weight Capacity	20000	N/A	lbs.	Reference Only. To be verified upon design of new facility
4	Platform Leveling Weight Capacity	35000	N/A	lbs.	Reference Only. To be verified upon design of new facility
5	Access Door Height	144	N/A	in.	Reference Only. To be verified upon design of new facility. Worst case. Other doors can be lower in height.
6	Access Door Width	168	N/A	in.	Reference Only. To be verified upon design of new facility. Worst case. Other doors can be lower in width.
7	Tool Move-in Path Width & Height (Dock to Fab)	122	N/A	in.	Based on SEMI standard for maximum tool size, also considering turning radius.

Appendix Q: Design Parameters for Process Cooling Water

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
a	Temperature	60	± 2	° F	Temperature Spec is equivalent to approximately ~15.5 ± 1°C.
b	Conductivity	200	<	µS/cm	Equivalent to Resistivity Spec of > 0.02 MΩ/cm.
c	Temperature Delta	10	N/A	° F	
d	Supply Pressure (Fab level)	80	± 2	psig	Equivalent to 0.552 ± 0.0135 MPa.
e	Supply Pressure (Subfab level)	87	± 2	psig	Equivalent to 0.600 ± 0.0135 MPa.
f	Maximum Differential Pressure	65	N/A	psig	
g	Maximum Return Pressure	32	N/A	psig	Equivalent to 0.22 MPa.
h	Minimum Filtration	50	<	µm	

Appendix R: Design Parameters for Vacuum

Last Modified: 4/30/23

Version: 4

ID: ZHNTE2QEMQUW-1202974420-161

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Process Vacuum (PV)				
a	Pressure	-26	<	in. Hg (below atmospheric)	Site specific. Elevation may indicate otherwise
2	Arsenic Vacuum				Dedicated Arsenic vacuum system
a	Pressure	-10	<	in. Hg (below atmospheric)	Site specific. Elevation may indicate otherwise

Appendix S: Design Parameters for Ultra Pure Water/Reclaim Water

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Ultra Pure Water (UPW)				Detailed water quality (http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef80753474)
a	Temperature	72	± 2	° F	
b	Supply Pressure	55	± 5	psig	
c	Resistivity	> 18.0	N/A	Mohm-cm	
d	Maximum Particle/ml > 0.2um	< 0.3 @ 0.05	N/A	um	
e	Maximum Dissolved Silica	<0.05	N/A	ppb	
f	Maximum Total Silica	<0.05	N/A	ppb	
g	Maximum Oxygen	<5.0	N/A	ppb	
h	Maximum TOC	< 1.0	N/A	ppb	
i	Boron	<0.05	N/A	ppb	
2	Hot Ultra Pure Water (HUPW)				
a	Temperature	149	± 2	° F	
b	Supply Pressure	55	± 5	psig	
c	Resistivity	> 17.8	N/A	Mohm-cm	
d	Maximum Particle/ml > 0.2um	< 0.3 @ 0.05	N/A	um	
e	Maximum Dissolved Silica	<0.05	N/A	ppb	
f	Maximum Total Silica	<0.05	N/A	ppb	
g	Maximum Oxygen	<10.0	N/A	ppb	
h	Maximum TOC	< 1.0	N/A	ppb	

Appendix T: Design Parameters for Waste Drains

Section	Description	FOR GLOBAL BOD			Notes/Comments
		Value	Tolerance	Units	
1	Ammonia Waste Water				Organic compounds, Ammonia, other Nitrogen
a	Temperature (Max)	90	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
2	Concentrated Acid Wastewater				Sulfuric Acid drain, Phosphoric Acid drain
a	Temperature (Max)	122	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
3	Concentrated Copper Wastewater				Copper solutions
a	Temperature (Max)	90	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
4	IW (Industrial Wastewater)				Hydrochloric Acid, Sulfuric Acid
a	Temperature (Max)	<149	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
5	Dilute Copper Wastewater				Copper solutions
a	Temperature (Max)	90	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
6	HFW (Hydrofluoric Acid Wastewater)				Hydrofluoric Acid
a	Temperature (Max)	<149	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
7	DIRC (DI Reclaim)				Selected third rinses from FAB
a	Temperature (Max)	<149	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
8	Solvent Waste				PGME, PGMEA, IPA
a	Temperature (Max)	Ambient	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
9	NPR (Non-Potable Water Reuse)				Second and third rinses from fab
a	Temperature (Max)	<149	N/A	° F	
b	Pressure	Atmosphere	N/A	psig	
c	Maximum TOC	<10	N/A	ppm	
10	Slurry Waste	Site-specific	10	Slurry Waste	Site-specific